

MASON CITY, IA, USA

WMO: 725485

Lat: 43.154N

Lon: 93.327W

Elev: 1225

StdP: 14.06

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14940

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-6.5	-17.6	2.2	-10.7	-12.9	2.9	-5.9	34.4	18.0	31.6	13.5	10.8	320	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	90.1	74.6	87.3	73.2	84.5	71.5	78.0	86.1	76.0	83.4	74.0	81.7	13.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.6	140.1	83.1	73.5	130.4	81.2	71.5	121.6	79.0	42.4	86.5	40.2	83.3	38.4	82.1	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.6	25.7	23.2	DB	-18.8	94.9	5.6	2.6	-22.8	96.7	-26.1	98.3	-29.2	99.7	-33.3	101.6
			WB	-19.1	81.5	5.5	2.1	-23.1	83.1	-26.3	84.3	-29.4	85.5	-33.5	87.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	16.0	20.2	33.0	46.5	58.5	68.7	72.0	69.3	62.3	49.0	34.7	21.8
	DBStd	21.91	12.95	12.59	12.22	9.96	8.52	6.52	5.65	5.53	8.19	9.55	10.65	11.73
	HDD50	4117	1053	835	544	178	21	0	0	0	7	136	469	874
	HDD65	7525	1518	1255	994	559	237	36	9	21	146	502	909	1339
	CDD50	2706	0	0	15	73	285	561	682	598	376	106	10	0
	CDD65	638	0	0	0	3	36	147	226	154	65	7	0	0
	CDH74	6103	0	0	6	76	416	1385	2123	1323	676	97	1	0
	CDH80	1911	0	0	0	19	121	454	729	372	194	22	0	0
Wind	WSAvg	11.1	12.5	12.5	12.5	13.3	12.0	10.2	8.4	7.7	9.0	10.9	11.9	11.9
Precipitation	PrecAvg	34.0	0.9	0.9	2.0	3.2	4.5	4.6	4.6	4.0	3.4	2.3	1.6	1.1
	PrecMax	46.7	2.4	3.5	4.8	8.2	12.9	11.4	12.2	15.7	14.4	5.6	4.2	3.9
	PrecMin	17.1	0.1	0.0	0.4	0.8	0.8	0.4	0.6	0.9	0.5	0.1	0.0	0.2
	PrecStd	6.0	0.6	0.6	1.1	1.6	2.2	2.6	2.7	2.9	2.8	1.5	1.3	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.4	53.9	72.3	82.6	89.2	92.9	94.3	91.0	89.7	83.1	68.7	54.5
		MCWB	40.5	48.0	57.8	63.2	69.5	74.4	77.9	75.5	71.5	64.9	55.9	50.0
	2%	DB	39.7	45.3	63.5	74.6	83.0	88.7	90.3	87.4	85.3	75.9	61.5	46.0
		MCWB	36.1	41.7	55.2	58.6	64.9	72.3	76.8	75.5	70.6	61.7	53.1	42.1
	5%	DB	35.8	40.3	56.4	69.0	78.4	85.3	87.0	84.2	81.6	70.1	55.5	40.7
		MCWB	33.4	37.3	50.1	55.0	63.1	70.1	74.9	73.2	67.9	59.6	48.4	37.8
	10%	DB	33.1	36.3	49.9	63.8	74.2	81.9	84.0	81.2	77.6	65.1	50.3	36.5
		MCWB	31.6	34.0	45.3	52.4	61.0	68.3	72.9	71.2	65.8	56.1	44.8	34.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.2	48.6	61.4	65.1	72.6	78.0	81.8	79.8	75.8	69.4	58.9	51.2
		MCDB	46.7	52.6	69.5	77.6	84.9	88.4	90.2	87.4	84.6	77.0	65.9	55.0
	2%	WB	36.4	41.7	56.1	60.8	68.9	75.2	78.9	77.1	73.0	64.7	54.2	42.7
		MCDB	39.2	44.9	62.0	71.1	78.8	83.8	86.6	84.8	82.2	71.7	60.3	45.3
	5%	WB	33.8	37.3	50.5	57.4	66.1	73.0	76.9	75.0	70.6	61.1	49.4	37.8
		MCDB	35.5	40.0	55.9	66.8	74.5	82.0	84.4	81.7	77.7	68.0	54.8	40.5
	10%	WB	31.7	34.2	45.0	53.5	63.3	70.8	74.7	72.9	68.2	57.3	45.1	34.3
		MCDB	33.3	36.1	50.2	61.6	71.2	79.2	81.8	79.0	74.9	64.8	50.2	36.1
Mean Daily Temperature Range	5% DB	MDBR	15.8	16.2	17.5	21.8	21.4	20.6	20.8	21.0	23.9	21.8	18.1	15.0
		MCDBR	18.4	18.7	27.2	31.7	27.4	24.2	22.9	23.2	27.3	29.2	26.4	19.8
		MCWBR	15.5	15.3	18.1	17.8	13.6	11.4	11.6	12.4	13.7	16.4	17.6	15.9
	5% WB	MCDBR	16.8	17.6	25.0	27.4	22.7	21.5	20.2	20.7	22.9	23.0	24.0	18.5
		MCWBR	14.5	14.6	17.5	17.1	13.0	12.1	11.7	12.4	12.7	15.4	17.3	15.3
Clear-Sky Solar Irradiance	taub	0.271	0.290	0.314	0.365	0.395	0.417	0.427	0.408	0.377	0.331	0.292	0.268	
	taud	2.417	2.367	2.423	2.318	2.301	2.281	2.239	2.289	2.345	2.465	2.504	2.446	
	Ebn at Noon	278	289	295	285	279	272	267	269	269	271	268	268	
	Edh at Noon	24	30	32	39	41	42	43	39	35	27	22	21	
All-Sky Solar Radiation	RadAvg	576	907	1197	1481	1745	1944	2015	1710	1384	909	605	446	
	RadStd	61	85	120	148	130	152	100	118	96	119	50	32	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page